Evaluation of Workflow Specific Prompts Responses

To evaluate the ability of LLMs to address workflow-specific background questions, we provide a set of foundational and technical prompts (Prompts are in the other file) related to representative Galaxy workflows. The results of this evaluation are shown in the following table, and the results indicate that Gemini 2.5 Flash consistently provides the most accurate and comprehensive responses. For example, in response to *W2–P3* (*What is gene annotation, and how is it used to associate peaks with genes?*), Gemini offers a detailed explanation encompassing both functional annotation and genomic coordinate mapping, whereas GPT-4o and DeepSeek-V3 provide less insight into the mechanisms linking peaks to nearby genes. Similarly, for *W5–P2* (*What is Bakta for bacterial genome annotation?*), Gemini elaborates on database usage and output formats, while the other models describe only its general purpose.

GPT-4o, although less comprehensive, stands out for its clarity and ease of understanding, making it particularly suitable for beginners. For example, its explanation of *W1–P1* (*What are nucleotides, chromosomes, exons, and SNPs?*) is concise, well-structured, and accessible to users with minimal background in genomics. DeepSeek-V3, while generally accurate, shows inconsistencies in depth and completeness. In *W3–P6* (*What does it mean for genes on opposite strands to overlap?*), DeepSeek fails to address key implications of antisense transcription and regulatory complexity that Gemini captures. These findings indicate that Gemini 2.5 Flash is best suited for users seeking technical rigor, GPT-4o excels in communication for broader audiences, and DeepSeek-V3 would benefit from greater consistency and contextual depth for bioinformatics education and support tasks.

Comparison of LLM Responses for Background Prompts in Galaxy Workflows

Criterion	**GPT-4o**	**Gemini 2.5 Flash**	**DeepSeek-V3**
Correctness	✔ Generally accurate; some minor conflation in technical distinctions	✔ Highly accurate and aligned with academic definitions	✔ Mostly correct but occasionally vague or incomplete
Clarity	✔ Very clear and accessible to beginners	⚠ Clear but often formal or verbose	⚠ Mixed clarity; some explanations too shallow or ambiguous
Completeness	⚠ Basic coverage; misses deeper insights	✔ Comprehensive, includes biological relevance and technical context	⚠ Omits some intermediate concepts or format distinctions
Conciseness	✔ Concise and to the point	⚠ Informative but often lengthy	✔ Generally concise but may lack depth
Terminology	✔ Uses common and understandable terms	✔ Uses precise and domain-specific vocabulary	⚠ Uses general terms; sometimes lacks specificity
Mistakes/Errors	Minor blending of concepts in definitions	Occasionally overuses jargon without explanation	Misses important context (e.g., gene–peak associations)
Overall Verdict	**Most user-friendly**	**Best academic rigor and depth**	**Needs improvement in consistency and detail**

To assess how well LLMs explain workflow-specific background concepts in Nextflow pipelines, we analyze their responses using representative Nextflow workflows. These prompts cover foundational biological and computational topics relevant to our workflows. The following table summarizes the comparative performance across the three models.

Gemini 2.5 Flash addresses complex topics such as metadata retrieval in *fetchngs*, reference genome handling in *phyloplace*, and bisulfite alignment nuances in *methylseq*, often incorporating tool-specific details, flags, and file structure references. For example, Gemini correctly explains the importance of ENA API calls for resolving sample metadata and provides detailed descriptions of output artifacts. Its responses are comprehensive, though sometimes overly verbose or formal, which may challenge novice users. GPT-4o, by contrast, prioritizes clarity and accessibility. It provides succinct, beginner-friendly explanations that are logically structured and easy to follow. For example, its explanation of the FASTQ format and the rationale for adapter trimming in RNA-seq workflows is precise and instructional. However, GPT-4o occasionally lacks depth in areas such as the resolution of experiment-level identifiers or the rationale behind multiple HMM profiles in *phyloplace*. DeepSeek-V3 delivers technically correct responses with a strong practical orientation. It excels in referencing command-line tools, directory structures, and usage examples. In the context of *fetchngs*, for instance, it correctly describes MD5 checksum validation and discusses the role of sample mapping fields. However, its answers often become list-heavy and miss some conceptual explanations, such as the biological rationale behind bin refinement in MAG analysis or how GAPPA supports phylogenetic visualization.

Comparison of LLM Responses for Background Prompts in Nextflow Workflows

Criterion	**GPT-4o**	**Gemini 2.5 Flash**	**DeepSeek-V3**
Correctness	✔ Generally correct with some minor oversights in tool-specific context	✔ Highly accurate and comprehensive across workflows	✔ Mostly accurate with detailed examples
Clarity	✔ Clear and easy to follow for most users	⚠ Slightly formal tone but readable	⚠ Occasionally dense and list-heavy
Completeness	⚠ Omits details like parameter flags and metadata resolution	✔ Covers technical tools, configurations, and context thoroughly	⚠ Strong examples, but some answers are tool-biased or partial
Terminology	✔ Balanced use of domain-relevant terms	✔ Advanced bioinformatics vocabulary used appropriately	⚠ Inconsistent: sometimes general, sometimes overly specific
Conciseness	✔ Well-balanced for instructional use	⚠ Verbose in longer answers	✔ Compact but requires interpretation
Mistakes/Errors	Minor: occasionally omits integration aspects or setup steps	Few: may overwhelm novices with detail	Minor: tool links are correct but lack rationale
Overall Verdict	**Best for clarity and beginners**	**Best for completeness and experts**	**Good for technical detail, less suited for guided explanation**

In summary, all three LLMs perform reasonably well in covering core concepts. Gemini 2.5 Flash is best suited for users seeking comprehensive and technically rigorous explanations, GPT-4o excels in readability and instructional clarity, and DeepSeek-V3 offers useful practical insights but requires refinement for conceptual completeness and consistency.